

Intro

Economy in the
medium-run

Inflation

Definition and measurement
Costs and benefits

Quantity theory

Phillips curve

Model
Modified PC

IS-LM model with
real interest rate
and risk premium

Real interest rate
Risk premium
Model

Summary

Lecture 6 Inflation

Economics B — Macroeconomics I

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Contents of the lecture

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Learning objectives and literature of the 6th lecture

- ▶ Learning objectives:
 - ▶ Economy in the medium-run
 - ▶ Importance of inflation
 - ▶ Explaining inflation: Quantity theory and Phillips curve
 - ▶ Extension of the IS-LM model to include the real interest rate and risk premium
- ▶ Literature: Textbook by Blanchard, chapters 8, 23.2, 23.3

Economy in the medium-run

- ▶ Lecture 5 introduced the medium-run: resource constraints
- ▶ In the medium-run, the available resources — for example, the production factor labor — limit potential output.
A persistently higher demand only leads to higher prices.
- ▶ We therefore analyzed the labor market in lecture 5 and now study the topic of inflation.

Motivation: Venezuela in the IS-LM model

- ▶ In the IS-LM model, an increase in the money supply leads to decreasing interest rates, which in turn stimulates the economy and increases GDP.
- ▶ Venezuela has done this excessively — with limited success:
 - ▶ The money supply in Venezuela grew so extremely in 2017/2018 that the bolívar fuerte became practically worthless, with over 1,000,000% inflation in 2018.
 - ▶ On August 20, 2018, the new bolívar soberano was introduced at an exchange rate of 1:100,000, but lost value just as quickly.
 - ▶ Real GDP growth declined by more than two thirds between 2013 and 2020. More than half of transactions now take place in dollars. Cryptocurrencies like the Petro are also used in some cases.
 - ▶ Data sources: CEIC Data (M2 Money Supply, Real GDP Growth, CPI)

Address by President Donald Trump at the WEF 2026

"Now, when they say the 'United States had a record quarter, it's unbelievable how well it's doing.' All the stocks crash because they say, 'Oh no, inflation, inflation, they're going to raise interest rates.' And they do, some of these stupid people like Powell. They raise interest rates. What they do is they stop you from being successful. It used to be when we had a great quarter, a great month, great earnings, great anything, any good news, the stock market went up. That's the way it's going to be, we got to do that again, because that's the way it should be."

Donald Trump, WEF 2026



WEF, 2026

Definition and measurement of inflation

- ▶ Inflation rate in 2025: $\pi_{2025} \equiv [P_{2025} - P_{2024}] / P_{2024}$
- ▶ Measurement usually in two ways:
 - ▶ Consumer price index (CPI) as a measure of inflation in consumption.
How does the price of an average basket of goods change?
 - ▶ GDP deflator = nominal GDP / real GDP
- ▶ CPI and GDP deflator differ — for example, in industrial and government consumption, in imports, or in the weighting of goods.

Problems with measuring inflation

- ▶ Measuring inflation is made more difficult by quality improvements and new products → Aghion et al. (*AER*, 2019)
- ▶ Do we measure inflation correctly? The phenomenon of 'shrinkflation':



Source: 20 Minuten, May 4, 2019

Inflation measurement in Switzerland

CPI basket 2025 – weights of main groups

Data: FSO

Health care (15.6%)	Other goods and services (5.8%)	Clothing and footwear (2.6%)	Alcoholic beverages and tobacco (2.5%)	Education (0.8%)
		Household goods and services (3.1%)	Communication (2.6%)	
Housing and energy (27.0%)	Restaurants and hotels (9.5%)		Recreation and culture (8.9%)	
	Transport (11.3%)		Food and non- alcoholic beverages (10.4%)	

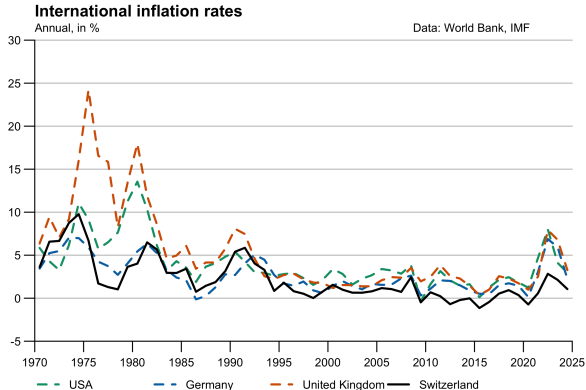
- For comparison: Food had a weight of 21% in the 1980s.
- Do all people consume goods in these proportions?

International comparisons

- ▶ International comparisons of inflation rates are complicated by the use of different basket weights.
- ▶ Example: Harmonized Index of Consumer Prices of the Eurozone, weighting of food and non-alcoholic beverages:
 - ▶ Austria 12.1%
 - ▶ Germany 13.1%
 - ▶ France 14.7%
 - ▶ Italy 18.1%
 - ▶ Spain 18.0%
 - ▶ Portugal 21.1%
 - ▶ Latvia 26.8%

Inflation since 1970

- ▶ From the mid-1980s, inflation was a limited problem.
- ▶ Temporary rise in inflation rates after Covid-19:



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Real interest rate

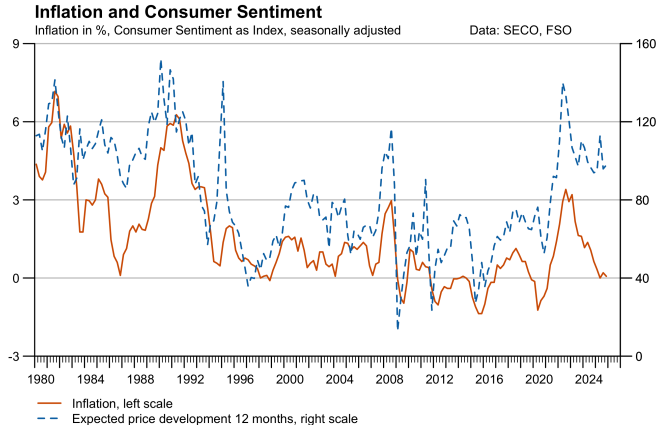
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Summary

Does the public have a good sense of inflation?

- The correlation between inflation and expected price developments is high (0.8).



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Costs of inflation

- ▶ High inflation is associated with uncertainty about inflation.
- ▶ Bad investments, redistribution from creditors to debtors
- ▶ Nominal contracts (bracket creep; wages and pensions, mortgages)
- ▶ Even constant inflation has costs: “menu costs”, “shoe leather costs”, costs of people turning to material assets (or a stable foreign currency such as the US dollar)
- ▶ Inflation reduces political re-election chances:
Many “do not like” inflation (behavioural economics argument).
- ▶ See also: Stable inflation is better than none, Avenir Suisse (2017)
- ▶ Examples of hyperinflation: Germany 1922/23, Zimbabwe 2006-08

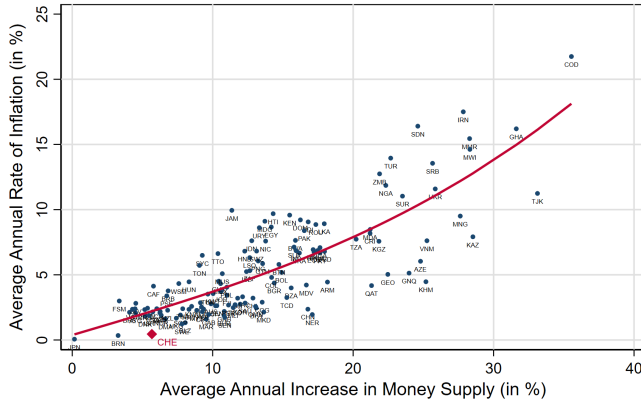


Benefits of inflation

- ▶ If (high) inflation is harmful to an economy, why don't central banks target a rate of 0%?
- ▶ The *initial* consequences of higher inflation are beneficial.
- ▶ Inflation increases the central bank's room for manoeuvre in stabilization policy, since nominal interest rates cannot become arbitrarily negative.
- ▶ Nominal wage cuts are hardly enforceable.
Inflation leads to real wage reductions.
- ▶ Seigniorage: State revenues from the creation of money.
This can be less distortionary than (other) taxes.

Money growth and inflation

- ▶ How does inflation arise?
- ▶ First explanation: The money supply increases (too fast).



Source: World Bank, FP.CPI.TOTL and FM.LBL.BMNY.CN
Time period from 2000 to 2015

Quantity theory of money — transaction version

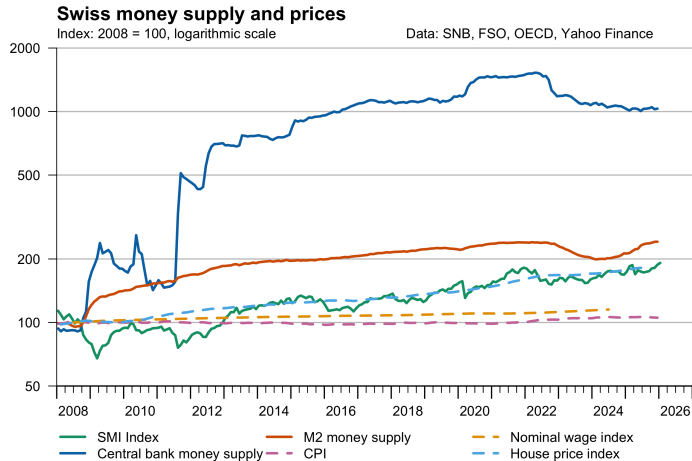
- ▶ “Inflation is always and everywhere a monetary phenomenon.”
Milton Friedman (1963), A Monetary History of the United States 1867–1960
- ▶ Equation of exchange: $M * V = P * T$
 M = money supply, V = velocity of money,
 P = price level, T = transactions
- ▶ Assumption 1: money supply M has no impact on T
- ▶ Assumption 2: V is constant in the medium-run, $\Delta V = 0$
- ▶ It follows: $\% \Delta M = \% \Delta P \quad \longrightarrow$ Direct relationship between M and P .
Money supply and price level proportional to each other
 $\longrightarrow$ Inflation is caused by growth of the money supply.

Quantity theory of money — income version

- ▶ The number of transactions is difficult to determine: replace T with real GDP (Y).
- ▶ The quantity equation is then: $M * V = P * Y$
 M = money supply, V = velocity of money,
 P = price level, Y = real GDP

Quantity theory of money — data from Switzerland

- How have money supply and prices developed in Switzerland?



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Did quantitative easing (QE) lead to inflation?

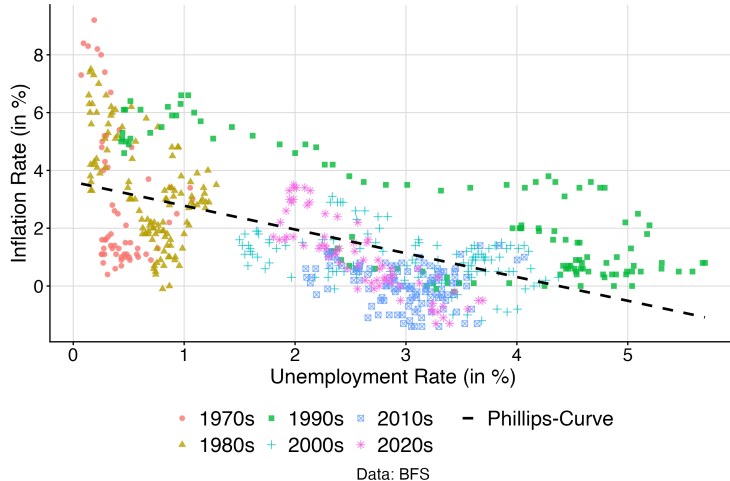
- ▶ In the normal case, money is created primarily by banks' lending. But the increase in M0 has been much larger than in M2. Most of the money does not end up in the real economy at all.
 - ▶ The velocity of circulation (V) is not constant.
- ▶ But the M2 money supply has also increased substantially. Why was there no CPI inflation until 2021?
- ▶ The inflation occurred primarily in financial markets and the real estate market, without being measured as inflation in the CPI. We primarily saw rising stock prices and real estate prices.

The Phillips curve as a second explanation

- ▶ The Phillips curve is named after the New Zealand-born statistician and economist William Phillips (1958).
 - ▶ “The Relation between Unemployment and the Rate of Change of Money Wages in the United Kingdom, 1861–1957”, *Economica*, Vol. 25, 283–299
- ▶ It describes a negative empirical correlation between
 - ▶ wage changes (or price level changes) and
 - ▶ unemployment rate
- ▶ From the beginning, caution was advised in interpretation:
The correlation does not mean that, for example, strong wage increases lead to lower unemployment.

The Phillips curve in practice

- The postulated relationship in Swiss time series data:



The Phillips curve — idea

- ▶ Why might the empirical correlation arise?
- ▶ Idea: How do companies react to additional demand?
 - ▶ Overtime, more staff, subcontracting, raising prices
 - ▶ At high capacity utilization: primarily increase prices
- ▶ In a boom phase (low unemployment), wages tend to rise more strongly due to a better negotiating position for workers.
 - ▶ With higher wages the demand increases — how do companies react?
 - ▶ Companies pass on the increased wage costs in the form of higher prices.
- ▶ Time series data show that inflation (or wage growth) is higher in years with low unemployment.

Economic model to explain the Phillips curve

- ▶ From lecture 5:
 - ▶ Wage setting equation: $W = P^e * F(u, z)$
 - ▶ Price setting equation: $P = (1 + \mu) * W$ ($\rightarrow$ assumption $A = 1$)
- ▶ We now allow $P \neq P^e$. We replace W in the second equation with the expression of the first equation and get:
$$P = (1 + \mu) * P^e * F(u, z)$$
- ▶ Assume $F(u, z)$ has the following functional form: $F(u, z) = 1 - \alpha u + z$ where α describes how strongly wage negotiators react to u .
- ▶ Then we get: $P = P^e * (1 + \mu) * (1 - \alpha u + z)$
- ▶ By means of some transformations (cf. textbook pp. 193–194) we obtain:
$$\pi_t = \pi_t^e + (\mu + z) - \alpha u_t$$

Bonus slide: derivation

- Derivation of the last equation, slide 23: $P = P^e * (1 + \mu) * (1 - \alpha u + z)$
- With time indices and after dividing by P_{t-1} : $P_t/P_{t-1} = P_t^e/P_{t-1} * (1 + \mu) * (1 - \alpha u_t + z)$
- Expand the left side: $\frac{P_t - P_{t-1} + P_{t-1}}{P_{t-1}} = 1 + \frac{P_t - P_{t-1}}{P_{t-1}} = 1 + \pi_t$ note: $\pi_t \equiv \frac{P_t - P_{t-1}}{P_{t-1}}$
- Do the same with P_t^e so that: $(1 + \pi_t) = (1 + \pi_t^e)(1 + \mu)(1 - \alpha u_t + z)$
- We divide by $(1 + \pi_t^e)(1 + \mu)$: $\frac{(1 + \pi_t)}{(1 + \pi_t^e)(1 + \mu)} = 1 - \alpha u_t + z$
- As long as π_t , π_t^e and u_t are relatively small, the approximation holds:
 $1 + \pi_t - \pi_t^e - \mu = 1 - \alpha u_t + z$
- This can be rewritten as: $\pi_t = \pi_t^e + (\mu + z) - \alpha u_t$

Model interpretation

- ▶ How do we interpret $\pi_t = \pi_t^e + (\mu + z) - \alpha u_t$?
 - ▶ The equation describes a Phillips curve:
For given values for π^e , μ and z , a decrease in the unemployment rate u leads to an increase in inflation.
 - ▶ The Phillips curve shifts upwards when:
 - ▶ higher inflation is expected ($\pi^e \uparrow$)
 - ▶ the markup goes up ($\mu \uparrow$)
 - ▶ factors leading to higher wage demands are increasing ($z \uparrow$)
- ▶ Where do inflation expectations come from, how is π_t^e determined?

Anchored inflation expectations

- ▶ If the inflation rate is stable at a level π^* over several years and without a trend, rational individuals expect inflation of π^* for the next year, so that $\pi_t^e = \pi^*$.
- ▶ In this case, low unemployment leads to high inflation (provided that μ and z remain unchanged). → see previous slide
- ▶ It seems that there is a trade-off between inflation and unemployment.
 - ▶ “It seems to me that the German people – to put it bluntly – can tolerate a 5% price increase more easily than a 5% unemployment rate”.
Helmut Schmidt in the *Süddeutsche Zeitung*, July 28, 1972
- ▶ Can a government really control the unemployment rate by choosing an appropriate level of inflation?

Collapse of the Phillips curve

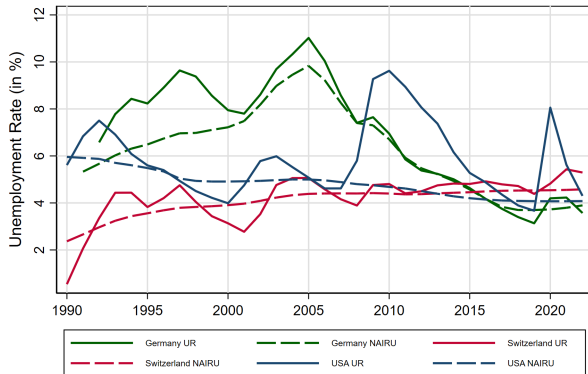
- ▶ Stagflation (= stagnation & inflation) in the 1970s:
The empirical correlation between inflation and unemployment collapses.
- ▶ In the medium-run, workers recognize that the inflation rate is not stable and adjust their expectations (π_t^e).
→ the Phillips curve is moving up.
- ▶ Reason: Wage setters adjust their expectations: $\pi_t^e = (1 - \theta)\pi^* + \theta\pi_{t-1}$, where $0 \leq \theta \leq 1$ indicates how strongly expectations for inflation in t depend on inflation in $t - 1$.
- ▶ Due to persistent inflation increases, θ rose to 1, so that the equation from slide 23 changes to: $\pi_t - \pi_{t-1} = (\mu + z) - \alpha u_t$
- ▶ The unemployment rate influences the *change in the inflation rate*.
→ modified (extended by expectations) Phillips curve

Non-accelerating inflation rate of unemployment (NAIRU)

- ▶ Let us look again at $\pi_t = \pi_t^e + (\mu + z) - \alpha u_t$
- ▶ Milton Friedman and Edmund Phelps pointed out in the late 1960s that wage setters did not permanently underestimate inflation.
So, if $\pi_t = \pi_t^e$, we get $\alpha u_t = \mu + z$.
- ▶ If we solve this for u_t , we get the natural rate of unemployment:
 $u = u_n = (\mu + z)/\alpha$
- ▶ If we plug $\alpha u_n = (\mu + z)$ into the modified Phillips curve, we get:
 $\pi_t - \pi_t^e = -\alpha(u_t - u_n)$
- ▶ The Phillips curve is the relationship between the change in the inflation rate and the deviation of the unemployment rate from the natural rate of unemployment.
- ▶ That is why u_n is also known as
non-accelerating inflation rate of unemployment (NAIRU).

The natural unemployment rate (NAIRU) in practice

- Medium-run tendency of the unemployment rate towards the NAIRU:



Source: OEDC Economic Outlook 109, May 2021

- What economic policy conclusions follow from this?

Extensions of the IS-LM model

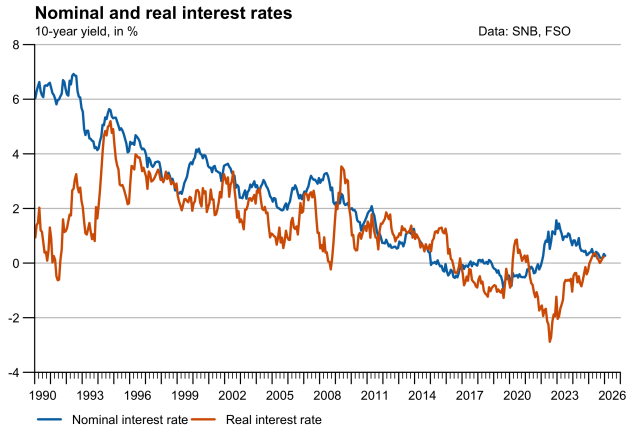
- ▶ The IS-LM model from lecture 4 is suitable for the analysis of the economy in the short-run.
- ▶ In the medium-run we must consider additional aspects:
 - ▶ Resource constraints (labor market): Equilibrium Y_n and u_n
 - ▶ Inflation: Prices change
- ▶ Therefore we extend the IS-LM model:
 - ▶ Distinction between nominal interest rate and real interest rate
 - ▶ Risk premiums
 - ▶ Resource constraints and inflation: Phillips curve (PC curve)
→ this follows in lecture 7

Extension of the IS-LM model: real interest rate

- ▶ In the IS-LM model for the short-run from lecture 4 there was only one interest rate controlled by the central bank ($\rightarrow$ LM curve).
- ▶ In the medium-run (i.e., with inflation) we must distinguish between nominal and real interest rates.
- ▶ Fisher equation: $i_t \approx r_t + \pi_{t+1}^e$
 - ▶ where $\pi_{t+1}^e = (P_{t+1}^e - P_t)/P_t$
 - ▶ The nominal interest rate in the current period t is the sum of the real interest rate and the expected inflation rate for the next period.
 - ▶ Exact formula: $(1 + i_t) = (1 + r_t) * P_{t+1}^e / P_t$

The real interest rate in Switzerland

- From Swiss data on nominal interest rates and realized inflation rates, real interest rates (ex post) can be calculated:

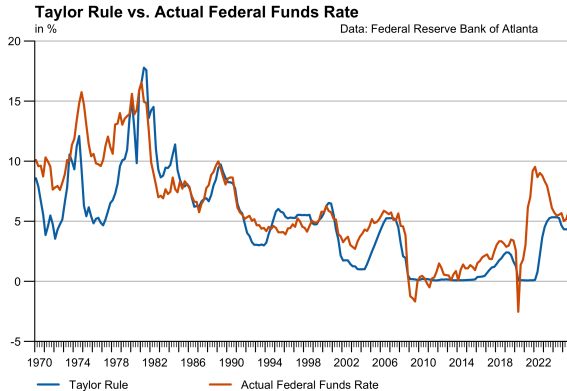


Monetary policy with the Taylor rule

- ▶ The central bank determines the real interest rate r indirectly via the nominal interest rate i .
- ▶ The central bank's objective function encompasses price stability and stable economic growth.
- ▶ One approach: the Taylor rule (or Orphanides rule in Europe)
- ▶ $i_t = r + \pi_t + a(\pi_t - \pi^*) + b(Y_t - Y_{n,t})$
 - ▶ Interpretation: The central bank chooses the interest rate i_t depending on the current inflation rate π_t (relative to the target rate π^*) and on the current GDP (relative to GDP on the long-term trend $Y_{n,t}$).
 - ▶ The parameters a and b determine how much value the central bank puts on price stability and economic growth, respectively.

Taylor rule in practice

- Originally presented in 1993 by John Taylor, modified over the years:



- What are the advantages of this predictability of monetary policy?

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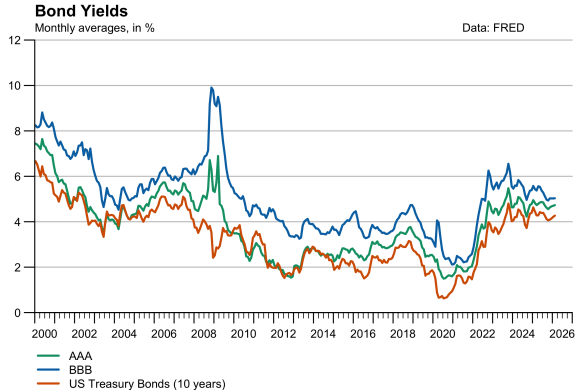
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Extension of the IS-LM model: risk premium

- ▶ Risk premiums: Since many bonds are risky, buyers of these bonds demand a risk premium.
- ▶ Let i be the nominal interest rate on safe bonds, x the risk premium, and p the probability of default.
- ▶ Then it holds for risk-neutral investors or lenders:
 - ▶ $(1 + i) = (1 - p) * (1 + i + x) + p * 0 \longrightarrow x = (1 + i)p / (1 - p)$
- ▶ The risk premium is determined by:
 - ▶ Here: probability of a payment default
 - ▶ In practice also: degree of risk aversion of the bond buyers

Extension of the IS-LM model: risk premium in the data

- The default risk of securities (*assets*) is assessed by rating agencies: AAA is the best rating, BBB is barely investment-grade.



IS-LM model with real interest rate and risk premium

- ▶ If we define the real interest rate as $r = i - \pi^e$ and the risk premium as x , we get new conditions for the IS-LM equilibrium:
- ▶ IS curve: $Y = C(Y - T) + I(Y, r + x) + G$
 - ▶ Investments depend on the real interest rate:
Companies must compensate financiers for inflation and risk.
- ▶ LM curve: $r = r_0 = i_0 - \pi^e$
 - ▶ The central bank now controls the real interest rate:
through the nominal interest rate and expectations about future inflation.
 - ▶ The real interest rate falls when the central bank reduces
the nominal interest rate (i) at given inflation expectations (π^e).

Summary and outlook

- ▶ Two explanations for inflation:
 - ▶ Quantity theory of money
 - ▶ Phillips curve
- ▶ Both approaches contribute to macroeconomic understanding, are discussed among economists due to the current situation.
- ▶ Over time, inflation expectations adjust to actual inflation. The state loses the possibility of 'trading-off' unemployment for inflation.
- ▶ In lecture 7 we will summarize these findings in the IS-LM-PC model. Preparation for this: slides and chapter 9 of the textbook

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Discussion and application to the current situation

- For practice, we present some questions on the lecture content during the lecture.
Please participate via QR code!